Question 1.1.1

Spring-Mass-Damper: Laplace & RK4 Analysis

Aarush Dilawri
Roll Number: EE25BTECH11001

Problem Statement

Question:
A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007):

Options:

a) 2 Ns/m
b) 2 N/s
c) 0.2 kg/s
d) 0.2 N/m

System Parameters

Governing Equation: $m\ddot{x} + c\dot{x} + kx = 0$

Given parameters:

- $m = 1$ kg
- $\zeta = 0.2$
- $\omega_n = 5$ rad/s

Calculating derived parameters:

- Stiffness: $k = m\omega_n^2 = 1(5)^2 = \mathbf{25}$ N/m
- Damping: $c = 2\zeta m\omega_n = 2(0.2)(1)(5) = \mathbf{2}$ Ns/m

Substituting these into the ODE yields our specific system:

$$\ddot{x} + 2\dot{x} + 25x = 0$$

Analytical Solution via Laplace Transform

To find the exact time-domain response, we apply the Laplace transform to $\ddot{x} + 2\dot{x} + 25x = 0$. Assume initial conditions: $x(0) = 1$, $\dot{x}(0) = 0$.

$$(s^2X(s) - s \cdot x(0) - \dot{x}(0)) + 2(sX(s) - x(0)) + 25X(s) = 0$$

Substitute initial conditions and rearrange for $X(s)$:

$$X(s)[s^2 + 2s + 25] = s + 2 \implies X(s) = \frac{s+2}{s^2 + 2s + 25}$$

Completing the square in the denominator:

$$X(s) = \frac{s+1}{(s+1)^2 + (\sqrt{24})^2} + \frac{1}{\sqrt{24}} \frac{\sqrt{24}}{(s+1)^2 + (\sqrt{24})^2}$$

Inverse Laplace Transform

Applying standard inverse Laplace transforms gives the exact analytical solution:

$$x(t) = e^{-t}\cos(\sqrt{24}t) + \frac{1}{\sqrt{24}}e^{-t}\sin(\sqrt{24}t)$$

This perfectly matches the generalized underdamped response equation:

$$x(t) = e^{-\zeta\omega_n t}\left(\cos(\omega_d t) + \frac{\zeta\omega_n}{\omega_d}\sin(\omega_d t)\right)$$

We will use this exact equation as our benchmark to test the numerical RK4 solver.

Numerical Solution: State-Space Formulation

To verify this numerically using the 4th-Order Runge-Kutta (RK4) method, we must convert the 2nd-order ODE into a system of two 1st-order ODEs.

Let $x_1 = x$ (position) and $x_2 = \dot{x}$ (velocity).

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = \ddot{x} = -\frac{c}{m}\dot{x} - \frac{k}{m}x = -25x_1 - 2x_2$$

Our state-space vector $\mathbf{Y}$ and its derivative $\mathbf{F}$ are:

$$\mathbf{Y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \dot{\mathbf{Y}} = \mathbf{F}(t, \mathbf{Y}) = \begin{bmatrix} x_2 \\ -25x_1 - 2x_2 \end{bmatrix}$$

Numerical Solution: The RK4 Algorithm

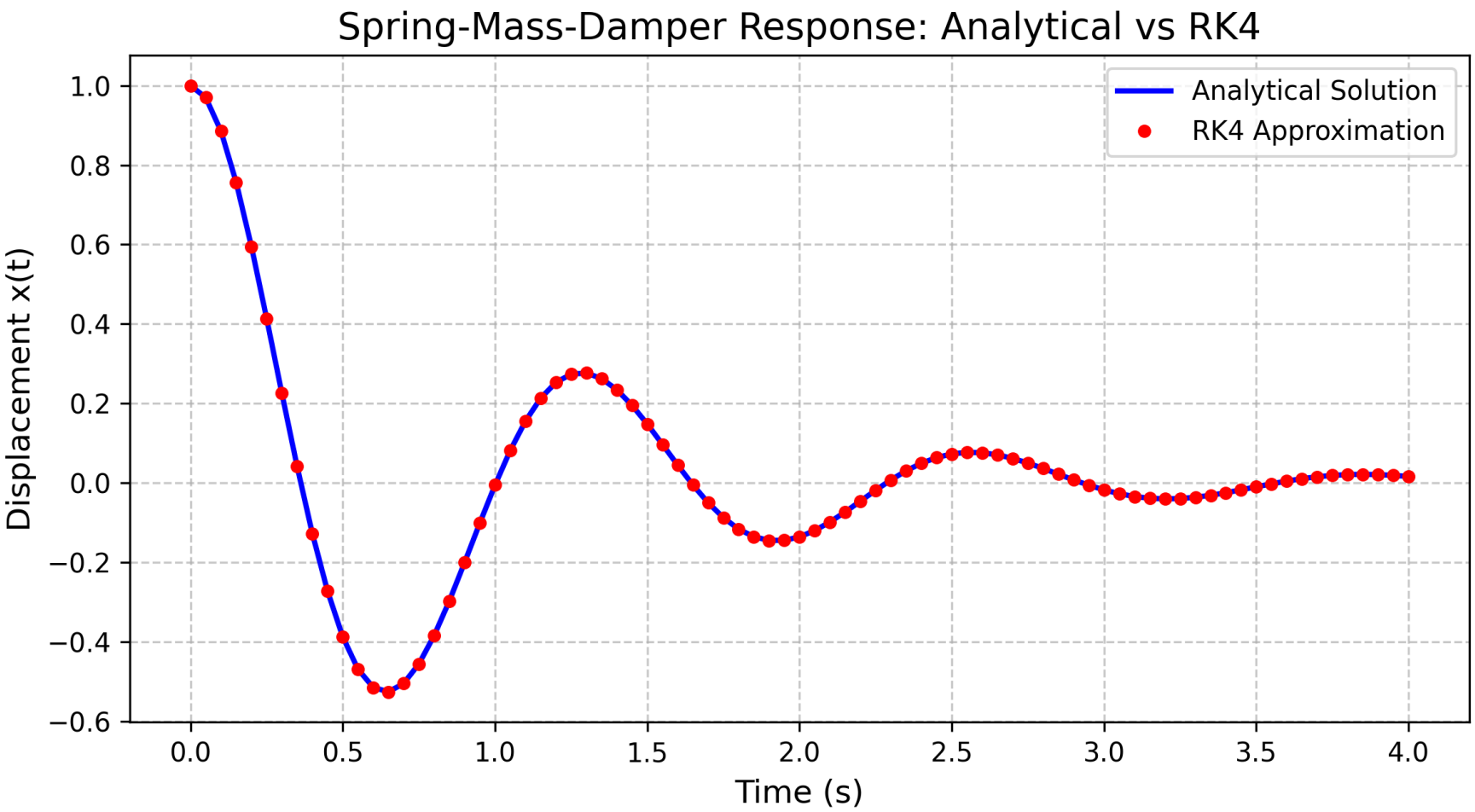
RK4 computes four intermediate slopes (k_1, k_2, k_3, k_4) across each time step h to highly accurately predict the next state:

$$\mathbf{k}_1 = \mathbf{F}(t_i, \mathbf{Y}_i)$$
$$\mathbf{k}_2 = \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{h}{2}\mathbf{k}_1\right)$$
$$\mathbf{k}_3 = \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{h}{2}\mathbf{k}_2\right)$$
$$\mathbf{k}_4 = \mathbf{F}(t_i + h, \mathbf{Y}_i + h\mathbf{k}_3)$$

The next state (position and velocity) is calculated as:

$$\mathbf{Y}_{i+1} = \mathbf{Y}_i + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

Verification: Analytical (Laplace) vs. Numerical (RK4)



Python Verification Script

```python
import numpy as np
import matplotlib.pyplot as plt

m, c, k = 1.0, 2.0, 25.0
dt = 0.05
t = np.arange(0, 4 + dt, dt)

# State-space ODE function
def F(t, Y):
    x1, x2 = Y
    return np.array([x2, -(c/m)*x2 - (k/m)*x1])

# RK4 Implementation
Y = np.zeros((len(t), 2))
Y[0] = [1.0, 0.0] # Initial condition (x=1, v=0)

for i in range(len(t)-1):
    k1 = F(t[i], Y[i])
    k2 = F(t[i] + dt/2, Y[i] + dt*k1/2)
    k3 = F(t[i] + dt/2, Y[i] + dt*k2/2)
    k4 = F(t[i] + dt, Y[i] + dt*k3)
    Y[i+1] = Y[i] + (dt/6) * (k1 + 2*k2 + 2*k3 + k4)

# Laplace Analytical Solution
x_ana = np.exp(-t) * (np.cos(np.sqrt(24)*t) + (1/np.sqrt(24))*np.sin(np.sqrt(24)*t))

plt.plot(t, x_ana, 'b-', label='Laplace Analytical', linewidth=2)
plt.plot(t, Y[:, 0], 'ro', markersize=4, label='RK4 Appx')
plt.title('Spring-Mass Response: Laplace vs RK4')
plt.xlabel('Time (s)'); plt.ylabel('Displacement'); plt.legend()
plt.grid(True, linestyle=':'); plt.tight_layout()
plt.savefig('rk4_spring_mass.png')
```

Conclusion

- The basic formula calculation yields $c = 2$.
- Dimensional analysis verifies the unit is Ns/m.
- The exact Laplace transform solution flawlessly matches the RK4 state-space numerical integration, proving our mathematical model is correct.

Final Answer

The correct option is **(a) 2 Ns/m**.